



COMPUTER AIDED ANALYSIS AND DESIGN

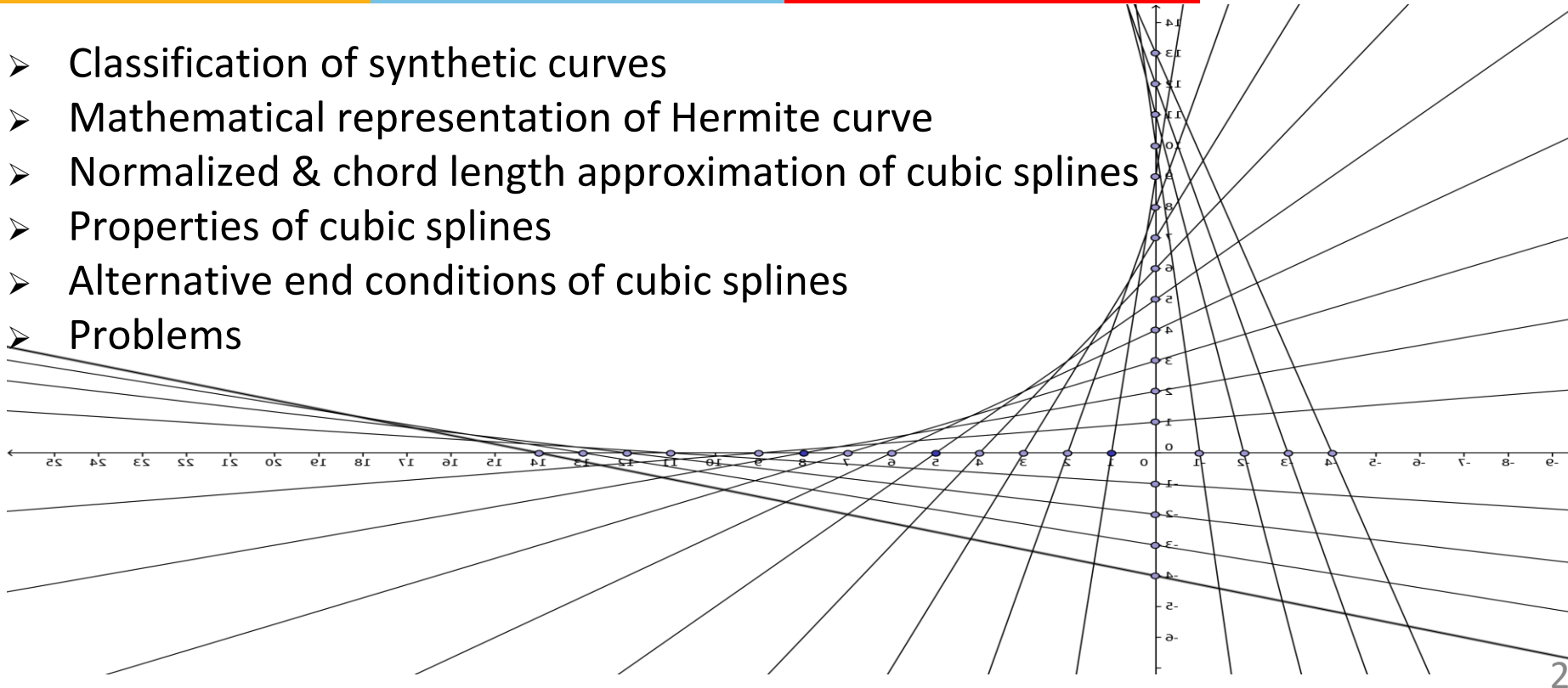
Course (DE ZG622) - Lecture 6

Prof. Ruchi More Kotecha

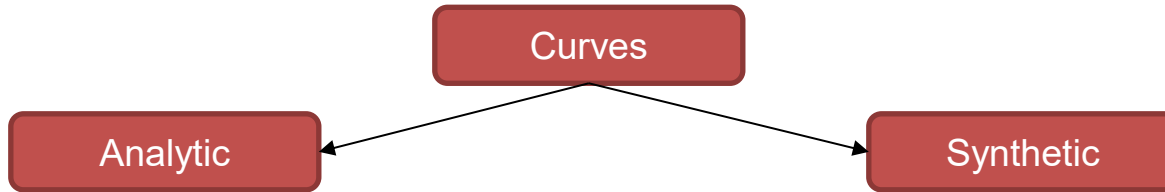
Session Outline



- Classification of synthetic curves
- Mathematical representation of Hermite curve
- Normalized & chord length approximation of cubic splines
- Properties of cubic splines
- Alternative end conditions of cubic splines
- Problems



Synthetic Curves



- Synthetic curves are the ones that are described by a set of data points called as control points.
- They offer flexibility in shape modeling and smoothness
- Eg. Splines, Bezier and B-spline curves
- These type of curves are used to model complex curves and surfaces required in real life applications such as propeller blades, ship hull, car body design, aircraft fuselage design etc.



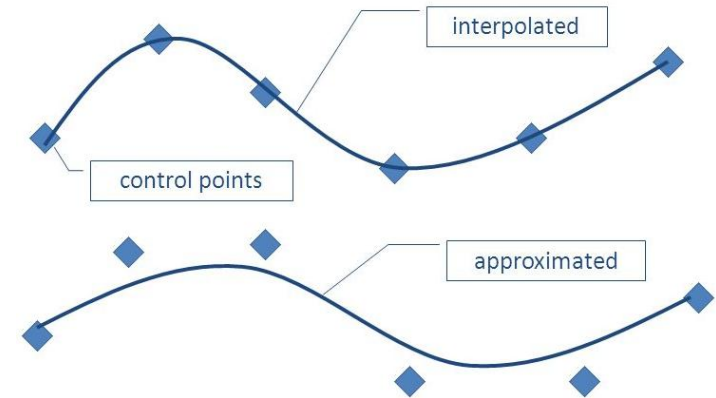
Need for Synthetic Curves



- When a curve is to be represented by set of measured data points with no upfront knowledge about equation of the curve. Here the curve must pass through the given set of points

OR

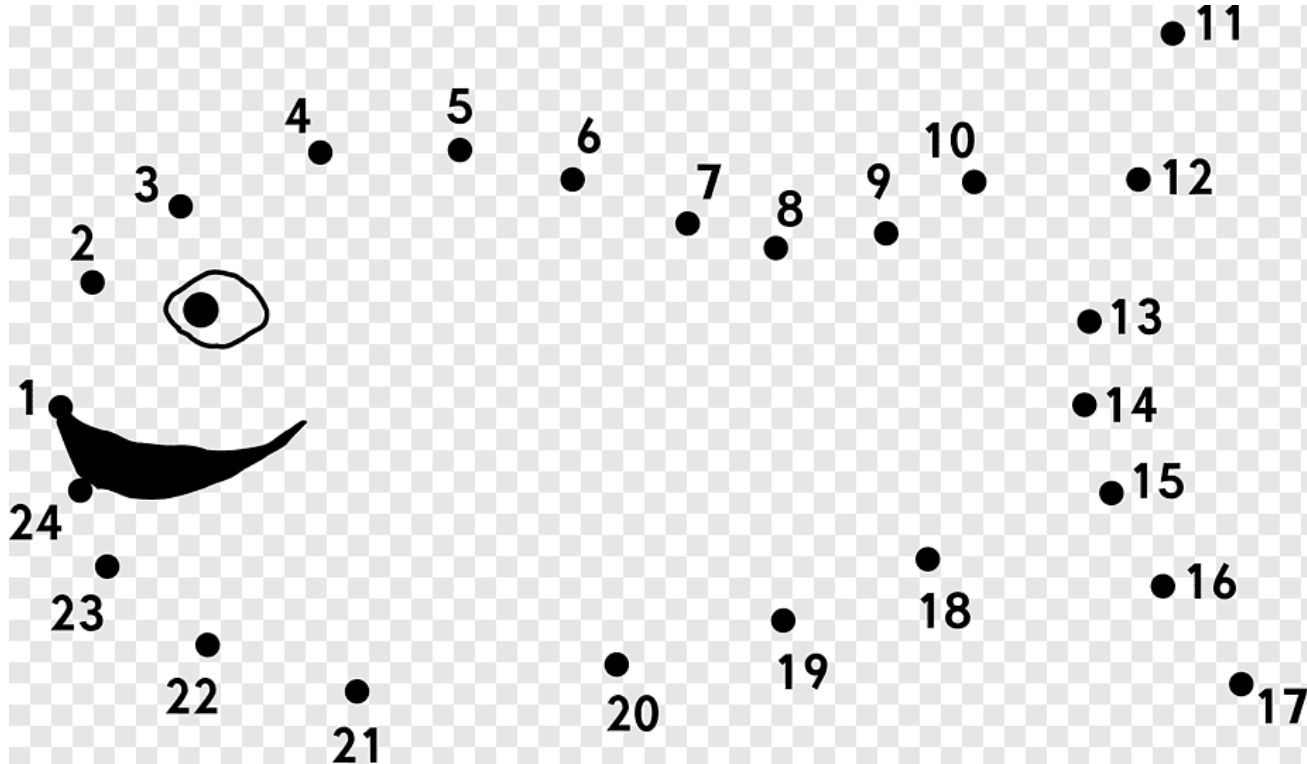
- When an existing curve needs to be manipulated to meet new design requirements with help of set of data points that can be used to bend, twist and change the shape of the curve. Here the curve may or may not pass through given set of points
- In both cases curve behaviour is governed by the data points and not by any governing equation. Hence these are synthetic curves.



Types of Curve Fitting:

- By Interpolation
- By Approximation

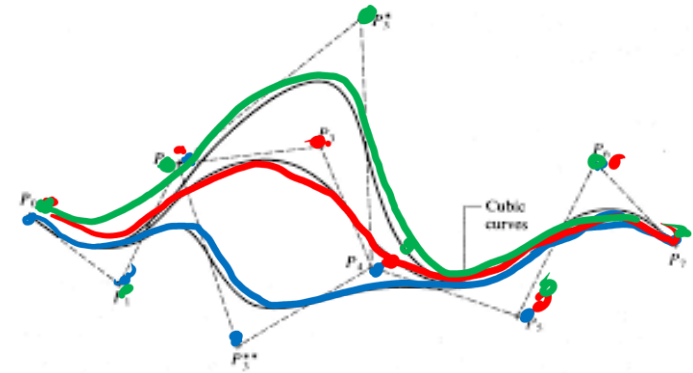
Curve Fitting - everybody's first assignment!



Characteristics of Synthetic Curves

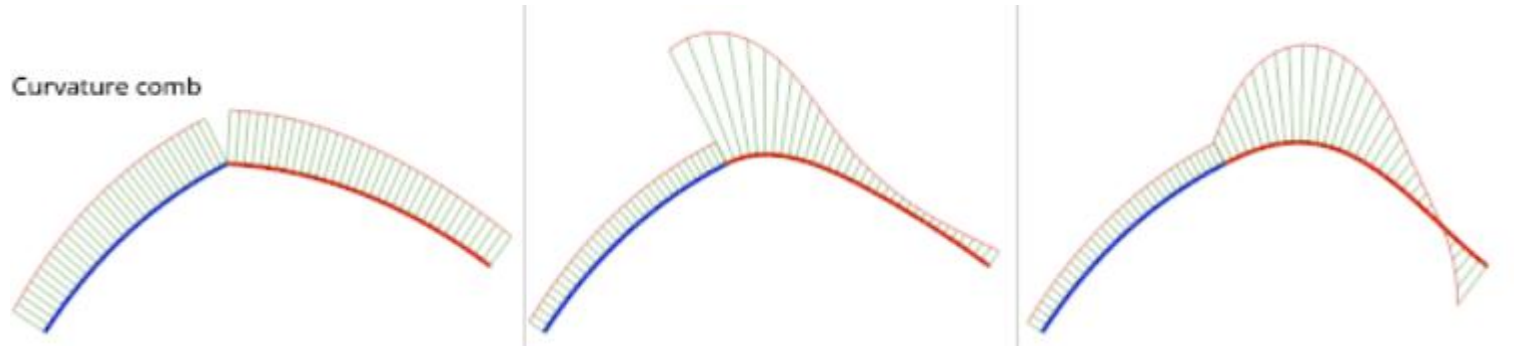


- A synthetic curve is a curve representation used in design, particularly for curved boundaries and surfaces. It is created by manipulating data points (called control points) to create bends and sharp turns in the shape of the curve.
- Essentially it is independent of coordinate geometry (though points on the curve can be computed in terms of coordinates).
- Smoothness of the curve is the most critical requirement of a synthetic curve.
- Synthetic curves are mathematically represented by polynomials.



Parametric Continuity of Curves

- A complex curve may be composed of multiple curve segments joined together.
- Smoothness of the overall curve depends upon how they are connected at joints. This is dictated by continuity of the curve.



C⁰ Continuity:

- Position

C¹ Continuity:

- Position + Slope

C² Continuity:

- Position + Slope + Curvature

Parametric Continuity of Curves

linear
quadratic
cubic

1
2
3

x

- C0 continuity indicates zero order continuity which means end of previous is same as start point of next segment.
- C1 continuity indicates first order continuity which means tangent at the end point of previous curve segment is same as tangent at the start point of next segment.
- C2 continuity indicates second order continuity which means curvature at the end point of previous curve segment is same as that at the start point of next segment.
- For engineering design a curve should be at minimum C1 continuous.
- Cubic polynomial is minimum order of polynomial that offers C0, C1 and C2 continuity at joints.



C^0 Continuity – The curve is Continuous everywhere

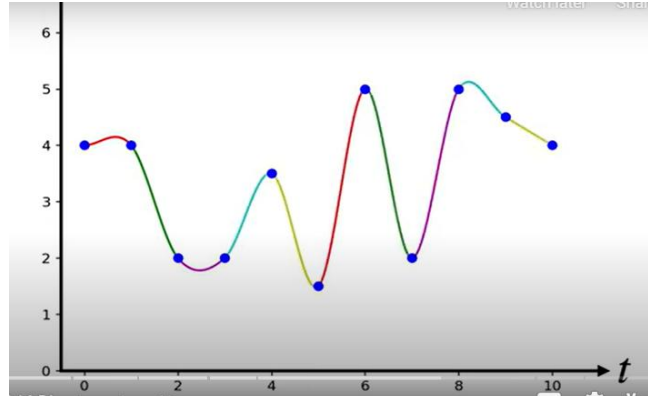
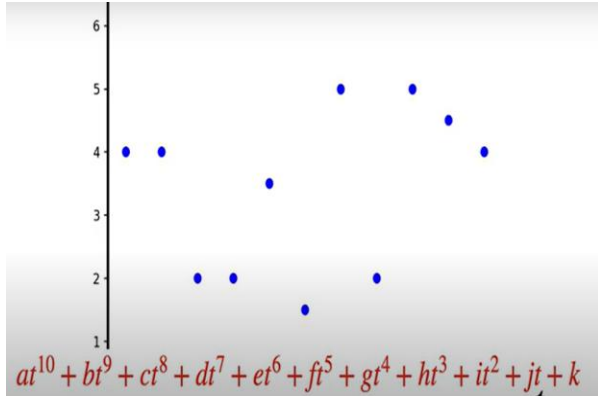


C^1 Continuity- Slope Continuity at the common point



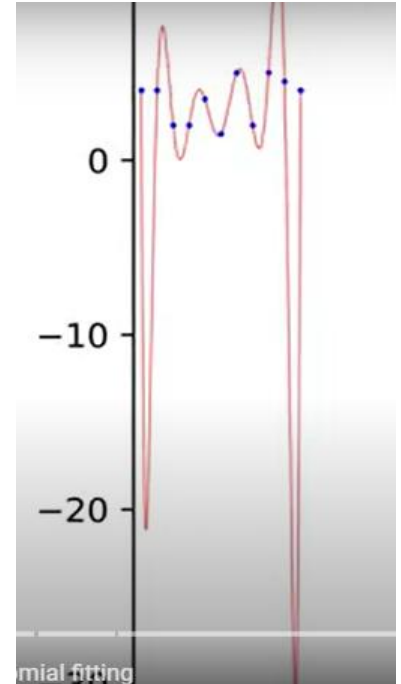
C^2 Continuity - Curvature continuity at the common point

Higher order vs Cubic (3rd order) polynomial curves



Cubic polynomial

Higher order polynomials tend to unduly oscillate about the control points and are computationally extensive/inconvenient, hence are not commonly used in CAD software.



10th order polynomial

Types of Synthetic Curves

➤ Cubic Splines or Hermite Curves

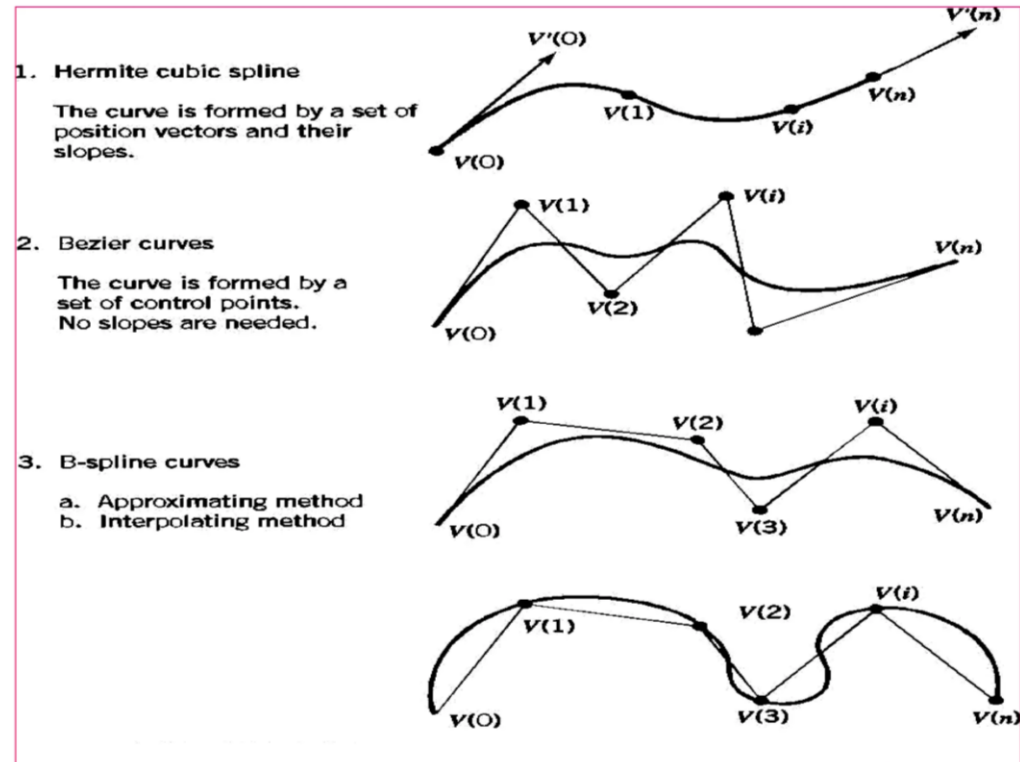
➤ (C0, C1 and C2 continuous)

➤ Bezier Curves

➤ (C0, C1, C2 continuous and beyond)

➤ B-Spline Curves

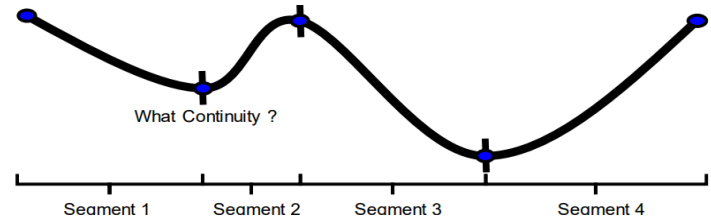
➤ (C0, C1 and C2 continuous)



Cubic Spline or Hermite Curve



- The name spline comes from the physical (instrument) spline which draftsmen use to produce curves
- Curve passes through all given set of points (control points) which is seen as interpolating piecewise polynomial
- These curves can be adjusted locally because each section is only dependent on its endpoint constraints.
- Shape of the Hermite curve can be determined by defining position and tangent vectors at the control points



Spline Tool



Parametric Representation of Cubic Spline

- A general cubic polynomial can be represented by

- $y = Ax^3 + Bx^2 + Cx + D$



- General parametric equation of cubic spline is given by

- $P(u) = \sum_{i=0}^3 C_i u^i \quad 0 \leq u \leq 1 \quad \text{----- (1)}$

Where u is a parameter and C_i are the coefficients of the polynomial

$$P(u) = \sum_{i=0}^3 C_i u^i \quad 0 \leq u \leq 1$$

$$P'(u) = \sum_{i=0}^3 C_i \cdot i \cdot u^{i-1} \quad 0 \leq u \leq 1$$

- Within a segment parametric cubic spline connects two end points and uses a cubic equation.
- In order to determine 4 coefficients, 4 conditions are required which are 2 position vectors of the end points and 2 tangent vectors at these points.

Parametric Representation of Cubic Spline



$$P(u) = C_3u^3 + C_2u^2 + C_1u + C_0 \text{ ----- (1)}$$

That is in matrix form

$$\mathbf{P}(u) = \mathbf{U}^T \mathbf{C}$$

Where

$$\mathbf{U} = [u^3 \quad u^2 \quad u \quad 1]^T \text{ and } \mathbf{C} = [C_3 \quad C_2 \quad C_1 \quad C_0]^T$$

$$\mathbf{P}(u) = [x(u) \ y(u) \ z(u)]$$

In scalar form:

$$\begin{aligned} x(u) &= C_{3x}u^3 + C_{2x}u^2 + C_{1x}u + C_{0x} \\ y(u) &= C_{3y}u^3 + C_{2y}u^2 + C_{1y}u + C_{0y} \\ z(u) &= C_{3z}u^3 + C_{2z}u^2 + C_{1z}u + C_{0z} \end{aligned} \text{ ----- (2)}$$

We need to find coefficients C_3 to C_0 corresponding to x , y and z .

Parametric Representation of Cubic Spline



Also differentiating equation of $P(u)$ in equation (1) we get tangent vector at P given by

$$P'(u) = \sum_{i=0}^3 C_i i u^{i-1} \quad 0 \leq u \leq 1 \quad \text{----- (3)}$$

$$P(u) = C_3 \cdot 3 \cdot u^2 + C_2 \cdot 2 \cdot u + C_1 \cdot 1 \cdot u^0 + C_0 \cdot 0 \cdot u^{-1}$$

$$P(u) = C_3 u^3 + C_2 u^2 + C_1 u + C_0$$

$$P'(u) = 3C_3 u^2 + 2C_2 u + C_1$$

This also can be expressed in terms of $x'(u)$, $y'(u)$ and $z'(u)$ similar to that shown in equation (2)

Applying first two conditions at P_0 where $u = 0$ and at P_1 where $u = 1$ in equation (1) and then applying next two conditions at P'_0 where $u = 0$ and at P'_1 where $u = 1$ in equation (3), we get

$$P_0 = C_0 \quad u=0 \quad P(u)$$

$$P'_0 = C_1 \quad u=0 \quad P'(u)$$

$$P_1 = C_3 + C_2 + C_1 + C_0 \quad u=1 \quad P(u)$$

$$P'_1 = 3C_3 + 2C_2 + C_1 \quad u=1 \quad P'(u)$$

Parametric Representation of Cubic Spline



Solving for C_0, C_1, C_2 and C_3 ,

$$\begin{cases} C_0 = \mathbf{P}_0 \\ C_1 = \mathbf{P}'_0 \\ C_2 = 3(\mathbf{P}_1 - \mathbf{P}_0) - 2(\mathbf{P}'_0 - \mathbf{P}'_1) \\ C_3 = 2(\mathbf{P}_0 - \mathbf{P}_1) + \mathbf{P}'_0 + \mathbf{P}'_1 \end{cases}$$

Substituting C_0 to C_1 in $P(u)$, where $P(u) = C_3u^3 + C_2u^2 + C_1u + C_0$,
we get position vector of P as:

$$\begin{aligned} \checkmark \mathbf{P}(u) &= (2u^3 - 3u^2 + 1)\mathbf{P}_0 + (-2u^3 + 3u^2)\mathbf{P}_1 \\ &\quad + (u^3 - 2u^2 + u)\mathbf{P}'_0 + (u^3 - u^2)\mathbf{P}'_1, \quad 0 \leq u \leq 1 \\ \mathbf{P}'(u) & \end{aligned}$$

Parametric Representation of Cubic Spline



In matrix form, it can be written as:

$[P] = [F][G]$ where

$$[F] = [(2u^3 - 3u^2 + 1) \quad (-2u^3 + 3u^2) \quad (u^3 - 2u^2 + u) \quad (u^3 - u^2)]$$

$$[G] = [P_0 \ P_1 \ P'_0 \ P'_1]^T$$

- Elements in matrix F are called **blending functions** that determine the shape of the curve between P_0 and P_1 .
- Elements in matrix G represent the geometric information.
- Thus any point P on the Hermite curve can be expressed in terms of parameter u at P , position vectors P_0 and P_1 and tangent vectors P'_0 and P'_1

Parametric Representation of Cubic Spline



Matrix $[F] = [(2u^3 - 3u^2 + 1) (-2u^3 + 3u^2) (u^3 - 2u^2 + u), (u^3 - u^2)]$

can be written as

$$[F] = [U]^T [M_H]$$

where,

$$[U] = [u^3 \ u^2 \ u \ 1]$$

$$[M_H] = \begin{bmatrix} 2 & -2 & 1 & 1 \\ -3 & 3 & -2 & -1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

M_H is called the **Hermite matrix**

Parametric Representation of Cubic Spline



Similarly differentiating equation of $P(u)$ we get tangent vector at P given by,

$$\mathbf{P}'(u) = (6u^2 - 6u)\mathbf{P}_0 + (-6u^2 + 6u)\mathbf{P}_1 \\ + (3u^2 - 4u + 1)\mathbf{P}'_0 + (3u^2 - 2u)\mathbf{P}'_1, \quad 0 \leq u \leq 1$$

Therefore, we can write this as

$$[\mathbf{P}'] = [\mathbf{B}'][\mathbf{G}]$$

Where,

$$[\mathbf{B}'] = [(6u^2 - 6u) \quad (-6u^2 + 6u) \quad (3u^2 - 4u + 1) \quad (3u^2 - 2u)]$$

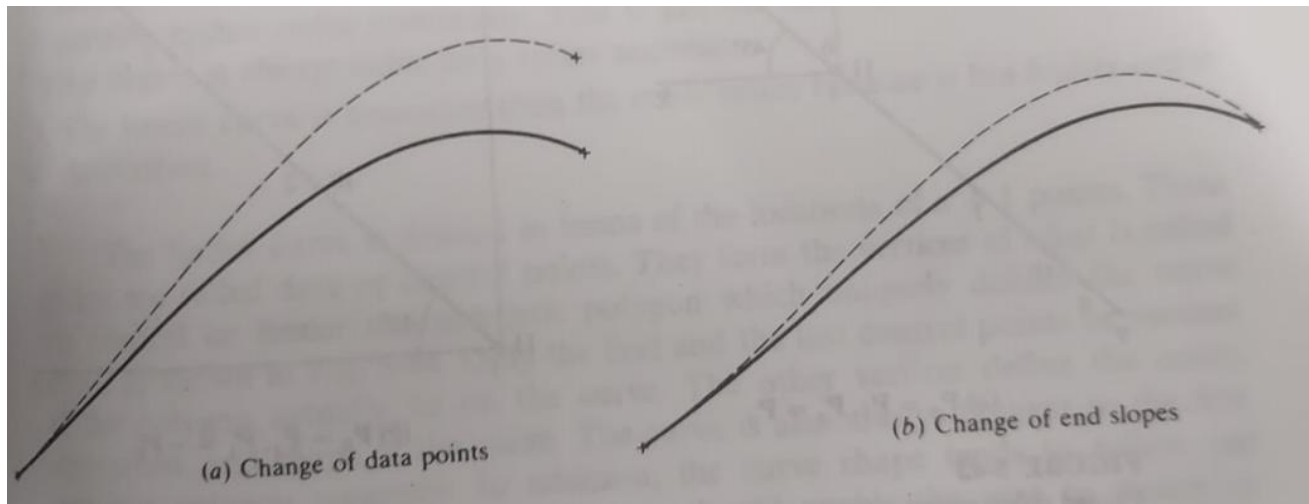
$$[\mathbf{G}] = [\mathbf{P}_0 \quad \mathbf{P}_1 \quad \mathbf{P}'_0 \quad \mathbf{P}'_1]^T$$

Elements in matrix $\mathbf{B}'$ are called **blending functions** for the tangent vector at point P on the cubic spline (Hermite curve)

Effect of Position and Tangent Vectors



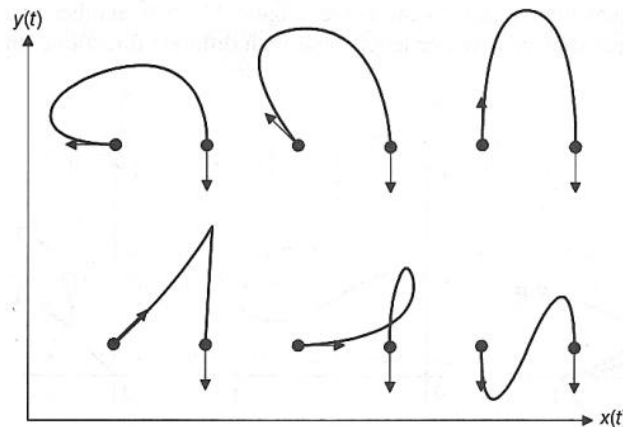
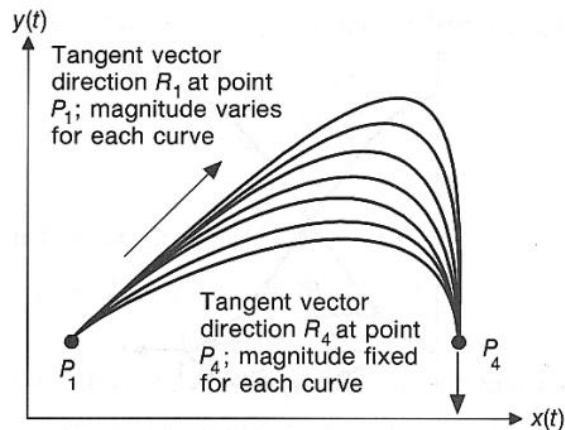
More importantly, the shape of the curve between P_0 and P_1 is governed by position vectors P_0 and P_1 , and tangent vectors P'_0 and P'_1 .



Effect of Tangent Vectors



Effect of tangent vectors (magnitude and direction) on the shape of Hermite curve with same end points.

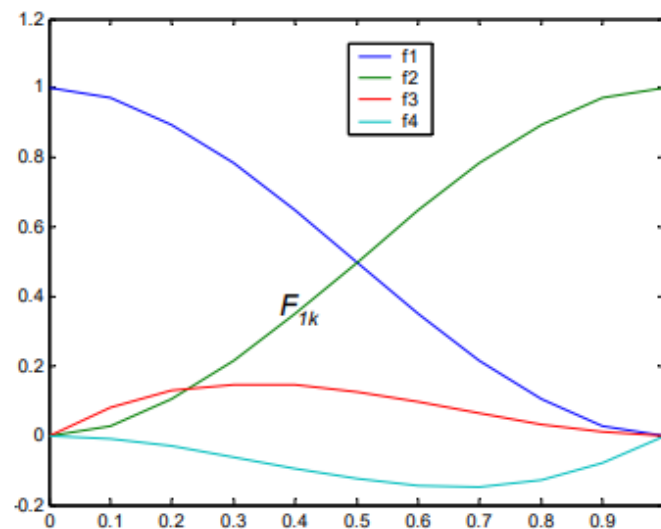


Blending Functions over Parametric Range



- F_1 is 1 at $u = 0$ and 0 at $u=1$. This indicates that F_1 has major contribution to shape of the curve near point P_0 and least contribution near P_1 .
- On similar lines, F_2 has major contribution to shape of the curve near point P_1 and least contribution near P_0 .
- Both blending functions for tangent vectors are zero at $u=0$ and at $u=1$. So they govern the shape of the curve between $u= 0$ to 1 where F_3 is dominant around $u= 0.25$ and reduces thereafter whereas F_4 is dominant around $u = 0.75$.
- At any point in the span sum of all blending functions is always 1 .

➤ Blending functions



Properties & Limitations of Cubic Spline



Properties of Cubic Spline:

- A cubic spline or the Hermite curve is defined by 2 end points 2 tangent vectors.
- Tangent vector control the direction and steepness of the curve at the end points
- Hermite curve is continuous and smooth offering C0, C1 and C2 continuity
- The Hermite form is computationally simple, easy to evaluate and compute. This is especially advantageous in real-time applications and animations.

Limitations of Cubic Spline:

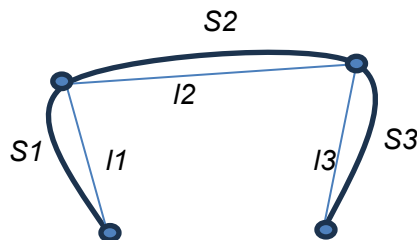
- The cubic splines interpolation algorithm does not work well for interpolation when the x values are large and have a large distance between them. Under these circumstances, cubic splines interpolation becomes very unstable making interpolations incorrect by many orders of magnitude. There is no local control support.
- The order of the curve is constant in spite of the number of data points.
- Physically, the values of blending function F_0 , F_1 , F_2 , and F_3 do not help in imagining the shape of the curve.

Normalized Cubic Splines



- If all the pieces of a composite cubic spline has parameters ranging between 0 to 1 then it is called the normalized cubic spline.
- These splines are fairly smooth and have uniformly distributed data points.
- They do not change the shape when control points undergo geometric transformations.

Chord Length Approximated Cubic Splines



$$u_1 = I_1 / (I_1 + I_2 + I_3)$$

$$u_2 = (I_1 + I_2) / (I_1 + I_2 + I_3)$$

$$u_3 = (I_1 + I_2 + I_3) / (I_1 + I_2 + I_3) = 1$$

For S1, $0 \leq u \leq u_1$

For S2, $u_1 \leq u \leq u_2$

For S3, $u_2 \leq u \leq 1$

- In chord length parameterization, the parameter (u) for each consecutive piece is chosen such that the distance between consecutive control points is proportional to the total chord length.
- This means that the parameter (u) is normalized over cumulative chord length from first to last control point.
- Chord length parameterization provides smoother curves.
- They typically do not self intersect.

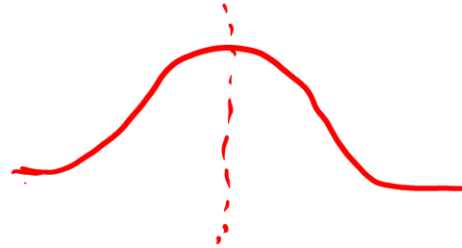
Alternative Cubic Spline End Conditions

Clamped Boundary Condition ✓

- This means that the first derivative (slope) of the spline at the endpoints is specified (known).
- Clamped conditions are used when the spline is desired to have specific slopes at the ends.

Relaxed Boundary Condition ✓

- This condition does not strictly fix the derivatives at the endpoints. Instead, these conditions allow the spline to "relax" to a natural or free form at the boundaries.
- Mathematically, the double derivatives or the curvatures at the end points are zero.
- $P''(0) = P''(1) = 0$



Alternative Cubic Spline End Conditions

Cyclic End Conditions

- This requires both the tangent vector and the curvatures at the ends be equal. These conditions are useful to produce closed or repeating cubic curves
- Mathematically,
- $P'(0) = P'(1)$ and
- $P''(0) = P''(1)$

Anticyclic Condition

- Anticyclic boundary conditions prevent periodicity between the endpoints. This allows for a smooth curve that does not wrap around, making it suitable for applications where a natural end-to-end connection is not desired.
- For e.g. Anticyclic end condition is when second derivative or the curvatures at the end points are opposite to each other.
- $P''(0) = -P''(1)$

Practice Problem Problems

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A cubic spline curve has start point P_0 (16, 0) and end point P_1 (3, 1). The tangent vector for end point P_0 is given by line joining P_0 and point P_2 (14, 8). The tangent vector for end P_1 is given by line joining P_1 and point P_2 . Plot the Hermite cubic curve and find its parametric equations.

$P' = \text{final} - \text{initial}$.

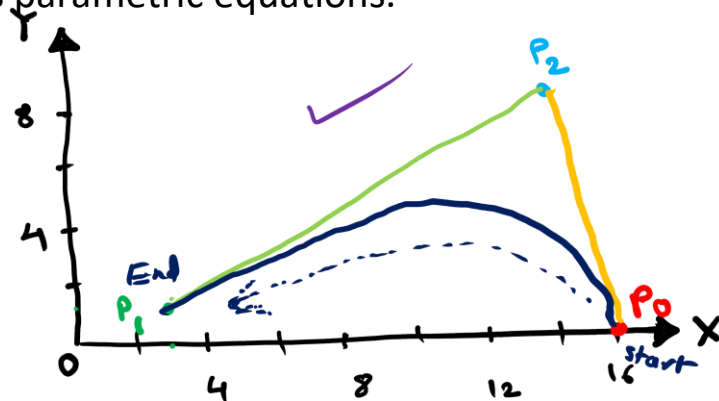
$$P'_0 = P_2 - P_0 = [14 \ 8] - [16 \ 0] = [-2 \ 8]$$

$$P'_1 = P_1 - P_2 = [3 \ 1] - [14 \ 8] = [-11 \ -7]$$

$$P(u) = [u^3 \ u^2 \ u \ 1] \begin{bmatrix} 2(P_0 - P_1) + P'_0 + P'_1 \\ -3(P_0 - P_1) - 2P'_0 - P'_1 \\ P'_0 \\ P_0 \end{bmatrix}$$

Approach # 1

substitute values
& find the answer.



Solution

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Approach #2

$$P(u) = [P_0 \ P_1 \ P_0' \ P_1'] [M_H] [U^T]$$
$$\begin{bmatrix} +2 & -3 & 0 & 1 \\ -2 & 3 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} u^3 \\ u^2 \\ u \\ 1 \end{bmatrix}$$

$$P_x(u) = [16 \ 3 \ -2 \ -11] [M_H] [U^T]$$

$$= [13 \ -24 \ -2 \ 16] [U^T]$$

$$= 13u^3 - 24u^2 - 2u + 16 \quad \text{--- (I) } \checkmark$$

$$P_y(u) = [0 \ 1 \ 8 \ -7] [M_H] [U^T]$$

$$= -u^3 - 6u^2 + 8u \quad \text{--- (II) } \checkmark$$

Solution



Solution



Solved Problems

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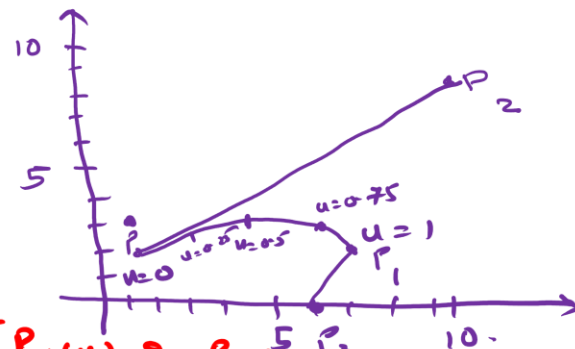
lead

A Hermite cubic curve has end points $P_0(1,3)$ and $P_1(7,2)$. The tangent vector at P_0 is defined by line joining P_0 and point $P_2(10,8)$ whereas tangent vector at P_1 is defined by line joining P_1 and $P_3(6,0)$. Compute the intermediate points on the curve at $u=.25, 0.5$ and 0.75

$$P'_0 = P_2 - P_0$$

$$P'_1 = P_3 - P_1$$

Substitute in equation of $P(u)$



substitute values in this eqⁿ to get $P_x(u)$ & $P_y(u)$.

$$P(u) = (2u^3 - 3u^2 + 1)P_0 + (-2u^3 + 3u^2)P_1 + (u^3 - 2u^2 + u)P'_0 + (u^3 - u^2)P'_1, \quad 0 \leq u \leq 1$$

$$P_x(u) - \textcircled{A}$$

$$P_y(u) - \textcircled{B}$$

$\textcircled{A}$ & $\textcircled{B}$ eq^{ns}.
will be in terms
of 'u'.

Now, substitute
 $u = 0.25, 0.5, 0.75$
& get values
of x & y .

$$P_0 \in (1, 3) \quad P_1(7, 2)$$

$$P_0' = P_2 - P_0 = \begin{bmatrix} 10 \\ 8 \end{bmatrix} - \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \end{bmatrix} \quad \begin{matrix} P_0' x \\ P_0' y \end{matrix}$$

$$P_1' = P_3 - P_1 = \begin{bmatrix} 6 \\ 2 \end{bmatrix} - \begin{bmatrix} 7 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \quad \begin{matrix} P_1' x \\ P_1' y \end{matrix}$$

$$P_{ux} = (2u^3 - 3u^2 + 1)P_{0x} + (-2u^3 + 3u^2)P_{1x} + (u^3 - 2u^2 + u)P_{0x}' + (u^3 - u^2)P_{1x}' \quad \textcircled{1}$$

Similarly

$$P_{uy} = (2u^3 - 3u^2 + 1)P_{0y} + (-2u^3 + 3u^2)P_{1y} + (u^3 - 2u^2 + u)P_{0y}' + (u^3 - u^2)P_{1y}' \quad \textcircled{2}$$

So substituting $u = 0.5$ in above values of P_0, P_1, P_0' & P_1' .

$$P_{0.25x} = [(2 \times 0.25^3 - 3 \times 0.25^2 + 1) \times 1] + [(-2 \times 0.25^3 + 3 \times 0.25^2) \times 7] \\ + [(0.25^3 - 2 \times 0.25^2 + 0.25) \times 9] + [(0.25^3 - 0.25^2) \times (-1)]$$

$$= [(0.03125 - 0.1875 + 1) \times 1] + [(-0.03125 + 0.1875) \times 7] \\ + [(0.015625 - 0.125 + 0.25) \times 9] + [(0.015625 - 0.0625) \times (-1)] \\ = [(0.84375) \times 1] + [0.15625 \times 7] + [0.140625 \times 9] + [-0.04875 \times 1]$$

$$= 0.84375 + 1.09375 + 1.265625 + 0.04875 \\ = 3.251875$$

Similarly calculate $P_{0.25y}$ that will give $P_{0.25}$

Similarly substitute $u = 0.5$ & 0.75 in above equations $\textcircled{1}$ & $\textcircled{2}$

Practice Problems

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#1) Find the mid point of a Hermite cubic spline with the two end points as $(1,1)$ & $(6,5)$ & corresponding tangent vectors as $(0,4)$ & $(4,0)$

Find equation of Hermite cubic spline and blending functions which passes through $(1,1)$ and $(3,2)$ and tangent at $(1,1)$ which defines by line connecting $(1,1)$ to $(0,1)$ and tangents at $(3,2)$ defines by line connecting $(3,2)$ to $(4,1)$

Ex-2. A cubic spline curve having end point $P_0(1, 3)$ & $P_1(7, 2)$. The two control point $P_2(10, 8)$ & $P_3(6, 0)$ are oriental such that lines P_0, P_2 & P_1, P_3 tangent to curves. Plot five point the Hermite cubic curve.

Thank You



COMPUTER AIDED ANALYSIS AND DESIGN

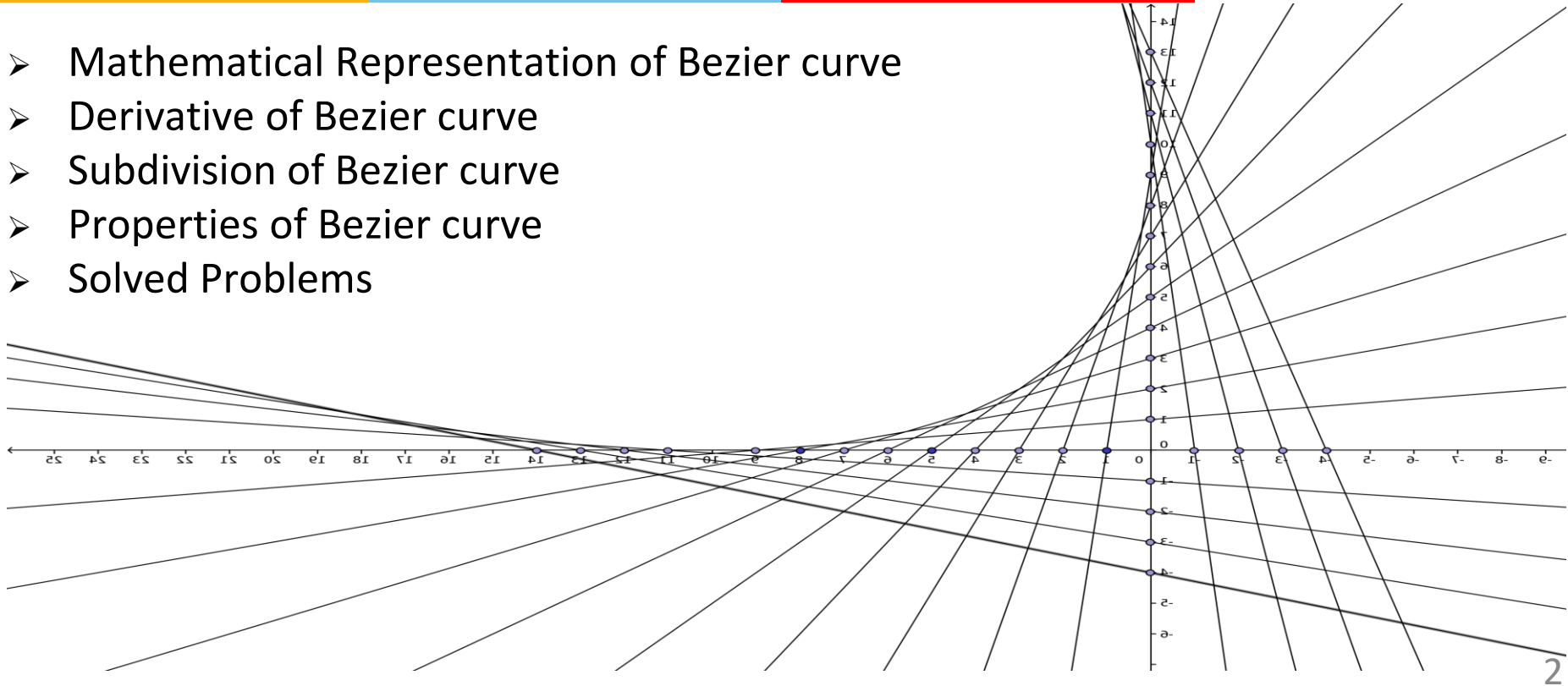
Course (DE ZG622) - Lecture 7

Prof. Ruchi More Kotecha

Session Outline



- Mathematical Representation of Bezier curve
- Derivative of Bezier curve
- Subdivision of Bezier curve
- Properties of Bezier curve
- Solved Problems



Limitations of Hermite Curves

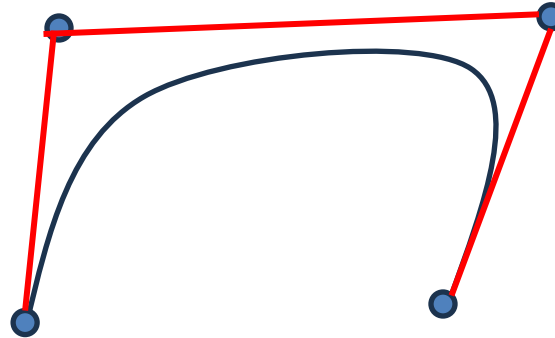
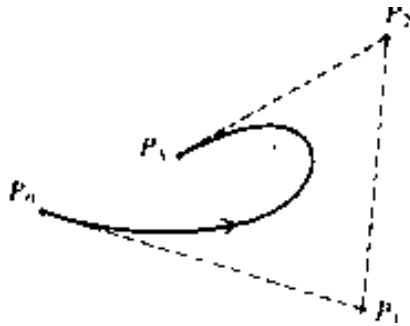


- First-order derivatives are required; it is not suitable for a designer to provide first-order derivatives.
- There is no local control support.
- The order of the curve is constant in spite of the number of data points.
- Physically, the values of blending function F_0 , F_1 , F_2 , and F_3 do not help in imagining the shape of the curve.

Bezier Curves



- Developed by P. Bezier of Renault motors, France. Used them in UNISURF software
- Independently developed by P. DeCasteljau of Citroen motors, France and used in their CAD system



Main Features of Bezier Curves



- The shape of a Bezier curve is defined by the controlled points only and no tangent vectors are needed. This provides more predictivity of resulting curve from given control points
- Degree of the Bezier curve is defined by the number of control points. Bezier curve of degree n is defined by $n+1$ control points
- Bezier curve is smoother than cubic spline because it can have higher order derivatives

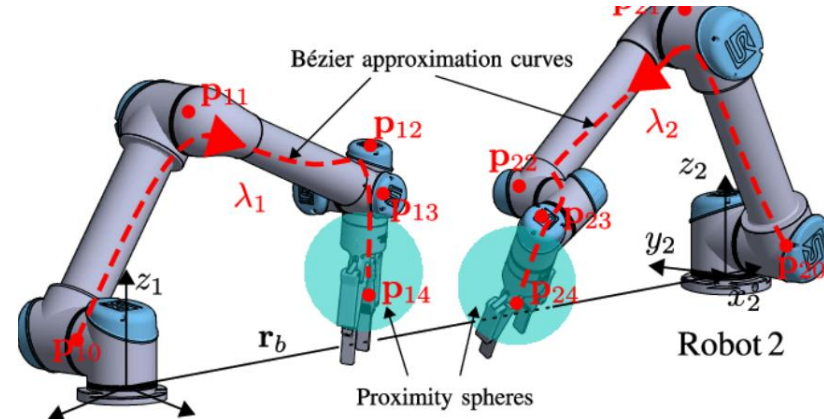
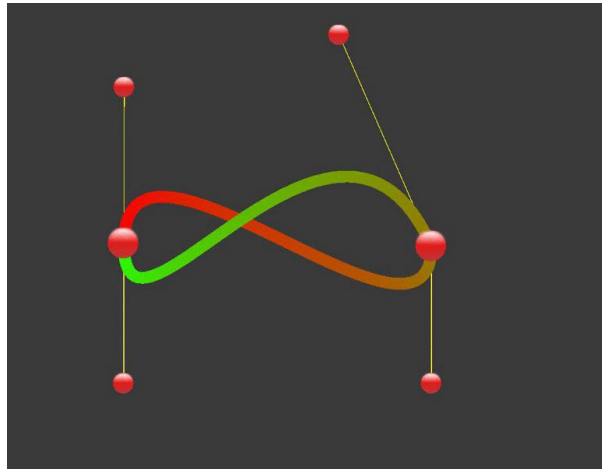
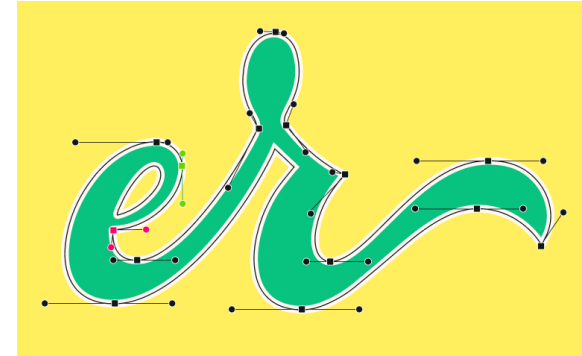
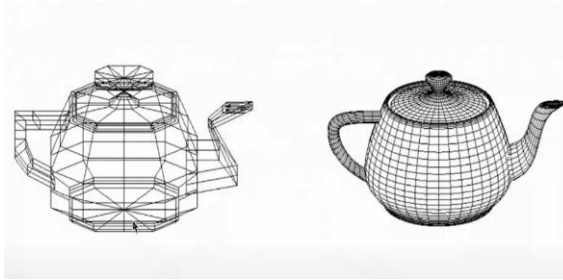
Applications of Bezier Curve

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- Computer graphics
- Fonts
- 3D animation
- Robotics

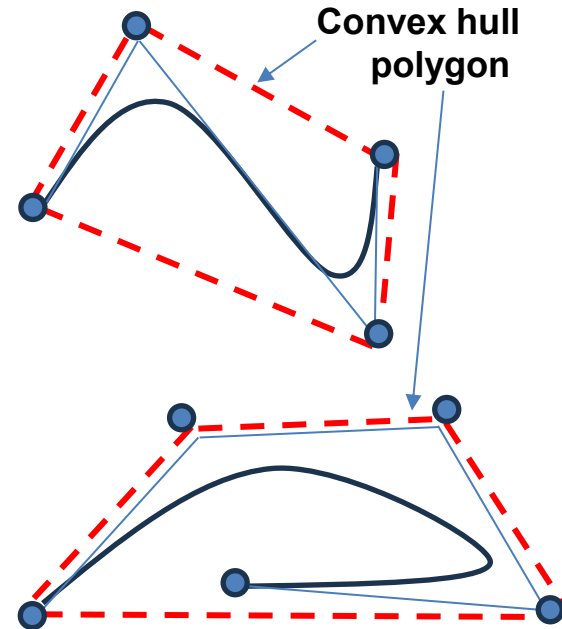
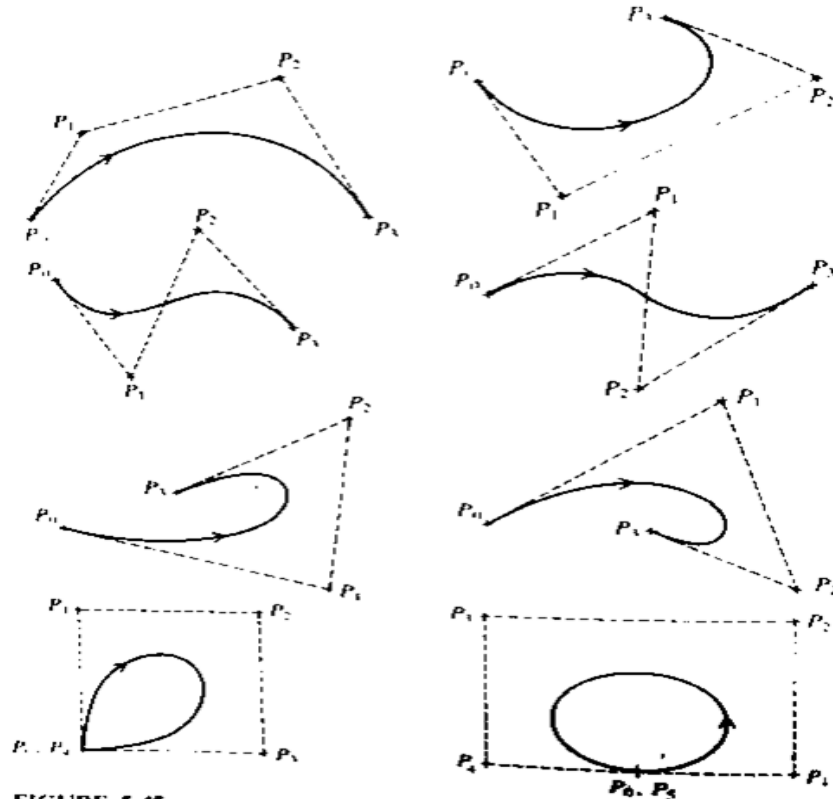


Properties of Bezier Curve



- Only the first and the last control points definitely lie on the curve.
- Other control points mainly define the order, derivative and shape of the curve.
- The curve is always tangent to the first and the last segment of the polygon formed by control points.
- Curve tends to follow the polygon shape.
- Change in the order of control points changes the shape of the curve.
- Bezier curve always remains in the convex hull of the defining control points polygon.

Bezier Curves and Convex Hulls



Mathematical Representation of Bezier Curve



$$P(u) = \sum_{i=0}^n P_i B_{i,n}(u) \quad 0 \leq u \leq 1$$

$P(u)$ is any point on Bezier curve and i is the control point, $(n+1)$ is total number of control points

$B_{i,n}(u)$ are called the **Bernstein Polynomials** or the **blending functions**

$$B_{i,n}(u) = C(n, i) u^i (1-u)^{n-i}$$

$C(n, i)$ is binomial coefficient given by $= \frac{n!}{i!(n-i)!}$

Substituting we get

$$P(u) = P_0(1-u)^n + P_1 C(n, 1)u(1-u)^{n-1} + P_2 C(n, 2)u^2(1-u)^{n-2} \\ + \dots + P_{n-1} C(n, n-1)u^{n-1}(1-u) + P_n u^n, \quad 0 \leq u \leq 1$$

Parametric Equation of Bezier Curve with 4 Control Points

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Eqⁿ for Bezier Curve: $P(u) = \sum_{i=0}^n P_i \cdot B_{i,n}(u)$

$$B_{i,n}(u) = C(n, i) u^i (1-u)^{n-i}$$

$n+1=4 \therefore n=3$

$C(n, i)$ is binomial coefficient given by = $\frac{n!}{i!(n-i)!}$

$$P(u) = \sum_{i=0}^n P_i \cdot \frac{n!}{i!(n-i)!} \cdot u^i (1-u)^{n-i} \quad \text{--- (I)}$$

$P(u) = P_0 \cdot B_{0,3}(u) + P_1 \cdot B_{1,3}(u) + P_2 \cdot B_{2,3}(u) + P_3 \cdot B_{3,3}(u)$ ←

$B_{0,3}(u) = \frac{3!}{0!(3-0)!} \cdot u^0 (1-u)^{3-0} = (1-u)^3 \quad \text{--- (A)}$

$B_{1,3}(u) = \frac{3!}{1!(3-1)!} \cdot u^1 (1-u)^{3-1} = 3 \cdot u \cdot (1-u)^2 \quad \text{--- (B)}$

$B_{2,3}(u) = \frac{3!}{2!(3-2)!} \cdot u^2 (1-u)^{3-2} = 3 \cdot u^2 \cdot (1-u) \quad \text{--- (C)}$

$B_{3,3}(u) = \frac{3!}{3!(3-3)!} \cdot u^3 (1-u)^{3-3} = u^3 \quad \text{--- (D)}$

Substitute terms A, B, C, D in (I)

$$P(u) = P_0 \cdot (1-u)^3 + P_1 \cdot 3 \cdot u \cdot (1-u)^2 + P_2 \cdot 3 \cdot u^2 \cdot (1-u) + P_3 \cdot u^3$$

Parametric Equation of Bezier Curve with 4 Control Points



$$B_{i,n}(u) = C(n, i) u^i (1-u)^{n-i}$$

$C(n, i)$ is binomial coefficient given by $= \frac{n!}{i!(n-i)!}$

substituting values of (a), (b), (c) & (d) in eqⁿ. I

$$P(u) = P_0 \cdot (1-u)^3 + P_1 \cdot 3u(1-u)^2 + P_2 \cdot 3 \cdot u^2(1-u) + P_3 u^3$$

Parametric Eqⁿ for Bezier Curves.

Parametric Equation of Bezier Curve with 4 Control Points



Here $n+1 = 4$, hence degree of the curve $n = 3$

Putting $n=3$ in previous equation and evaluating $C(n,i)$ for $n=3$ and i varying from 0 to 3 we get

$$P(u) = (1-u)^3 P_0 + 3u(1-u)^2 P_1 + 3u^2(1-u) P_2 + u^3 P_3$$

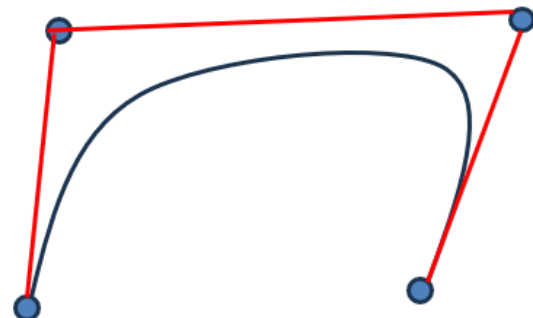
OR

$$[P] = [B][P]^T$$

$$[P] = \begin{bmatrix} \textcircled{A} & \textcircled{B} & \textcircled{C} & \textcircled{D} \end{bmatrix} [P_0 \ P_1 \ P_2 \ P_3]^T$$

Elements in $[B]$ are called **blending or basis functions**.

The polynomial is called **Bernstein Polynomial**.



Parametric Equation of Bezier Curve with 4 Control Points $\rightarrow P_0, P_1, P_2, P_4$.



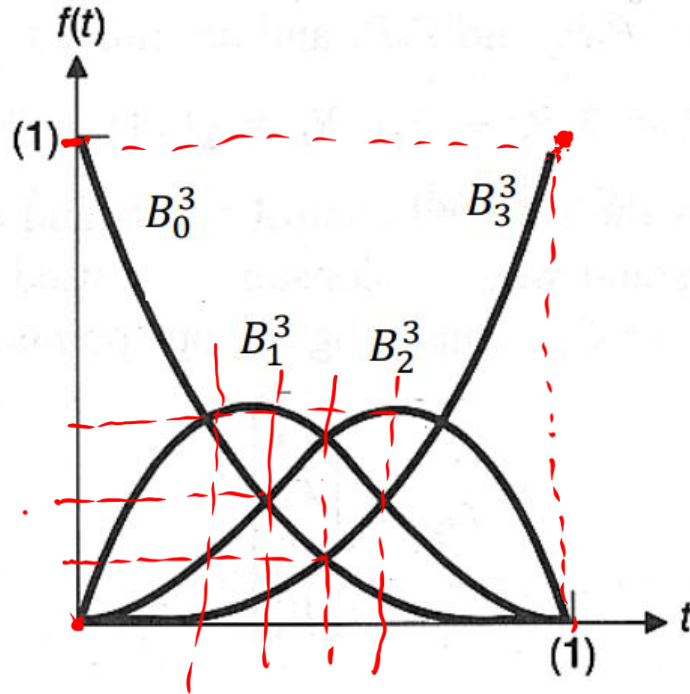
So x,y,z coordinates of any point P on the Bezier curve at parameter value u will be (This is nothing but parametric equation of Bezier curve)

$$P(u_x) = (1-u)^3 P_{0,x} + 3u(1-u)^2 P_{1,x} + 3u^2(1-u) P_{2,x} + u^3 P_{3,x} \quad \checkmark$$

$$P(u_y) = (1-u)^3 P_{0,y} + 3u(1-u)^2 P_{1,y} + 3u^2(1-u) P_{2,y} + u^3 P_{3,y} \quad \checkmark$$

$$P(u_z) = (1-u)^3 P_{0,z} + 3u(1-u)^2 P_{1,z} + 3u^2(1-u) P_{2,z} + u^3 P_{3,z} \quad \checkmark$$

Blending Functions of Bezier Curve



- The sum blending functions at any value of u is always 1.

Visual Representation of
Bezier curve:

<https://www.youtube.com/watch?v=aVwxzDHniEw>

More Properties of Bezier Curve



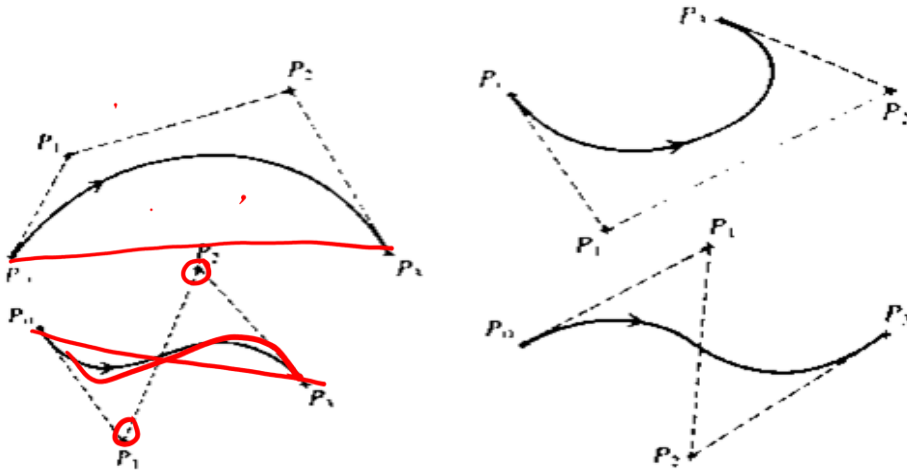
- Since Bezier curve is tangent to the first and the last segment, differentiating equation of $P(u)$ to get first derivative at point P_0 and P_1 we get
 - $P'(0) = n(P_1 - P_0)$
 - $P'(n) = n(P_n - P_{n-1})$
- Bezier curve is symmetric with respect to u and $(1-u)$. That is, even if the order of the control point is reversed the shape of the curve doesn't change.
- ✓ ➤ Each control point is most influential on the shape at $u = (i/n)$, that means for a four control points Bezier curve, control points P_0, P_1, P_2, P_3 are most influential at $0, 1/3, 2/3$ and 1 respectively.



More Properties of Bezier Curve



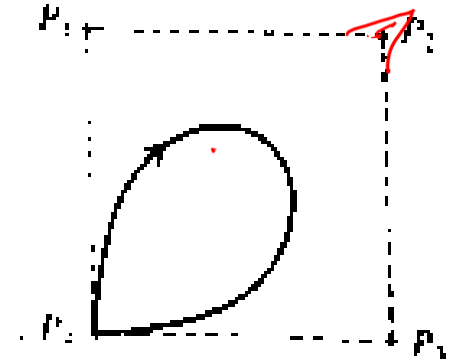
- Bezier curve exhibits the variation diminishing property.
- It means the curve do not oscillate about any straight line more often than the defining polygon.



Manipulating Shape of Bezier Curve



- The shape of the curve can be changed by changing the position of the control points.
- Adding multiple control points at a single position in space will add more weight to that point pulling Bezier curve towards it.
- A closed Bezier curve can be generated by closed corresponding convex polygon that making P_0 and P_n coincident.
- Bezier Curve Simulator (DIY)
- https://visualize-it.github.io/bezier_curves/simulation.html



Derivatives of Bezier Curve

$$B_{n,i}(u) = \frac{n!}{i!(n-i)!} u^i (1-u)^{n-i}$$

$$\frac{d}{du} B_{n,i}(u) = B'_{n,i}(u) = n(B_{n-1,i-1}(u) - B_{n-1,i}(u))$$

$$P'(u) = \sum_{i=0}^{n-1} B_{n-1,i}(u) \{n(P_{i+1} - P_i)\}$$

Substituting $Q_i = \{n(P_{i+1} - P_i)\}$

$$P'(u) = \sum_{i=0}^{n-1} B_{n-1,i}(u) Q_i$$

Thus **derivatives of Bezier curve are also Bezier curves.**

Derivatives of Bezier Curve



A simple example of taking the derivative of a Bezier curve can be shown using a cubic curve.

$$C_3(u) = \sum_{i=0}^3 B_{3,i}(u)P_i,$$

where $u \in [0, 1]$ and $B_{n,i} = \binom{n}{i}u^i(1-u)^{n-i}$ is the i -th Bernstein polynomial of degree n . P_i are the control points.

written out it is:

$$C_3(u) = (1-u)^3P_0 + 3(1-u)^2uP_1 + 3(1-u)u^2P_2 + u^3P_3$$

we can differentiate this with respect to u

$$\begin{aligned} \frac{\partial}{\partial u}C_3(u) = & -3(1-u)^2P_0 - 6(1-u)uP_1 + 3(1-u)^2P_1 - 3u^2P_2 + 6(1-u)uP_2 \\ & + 3u^2P_3. \end{aligned}$$

Derivatives of Bezier Curve



Now we can rearrange the terms

$$\frac{\partial}{\partial u} C_3(u) = 3 \left((1-u)^2(P_1 - P_0) + 2(1-u)u(P_2 - P_1) + u^2(P_3 - P_2) \right),$$

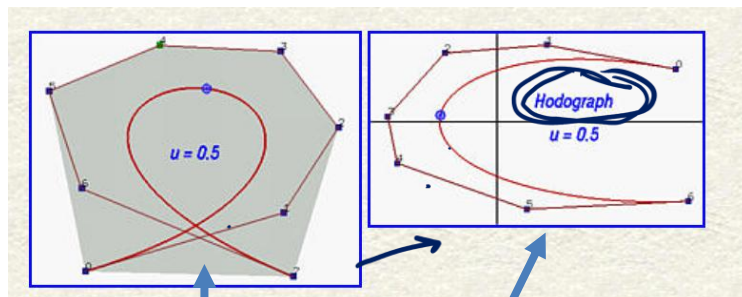
or

$$\frac{\partial}{\partial u} C_3(u) = 3 \left(\sum_{i=0}^2 B_{2,i}(u)(P_{i+1} - P_i) \right)$$

We obtain a quadratic B'ezier curve where the control points are the vectors between the initial control points, scaled by a factor 3. Therefore the derivative of a Bezier curve of degree n is a B'ezier curve of degree $n-1$, where the control points are difference between initial control points scaled by n . Therefore,

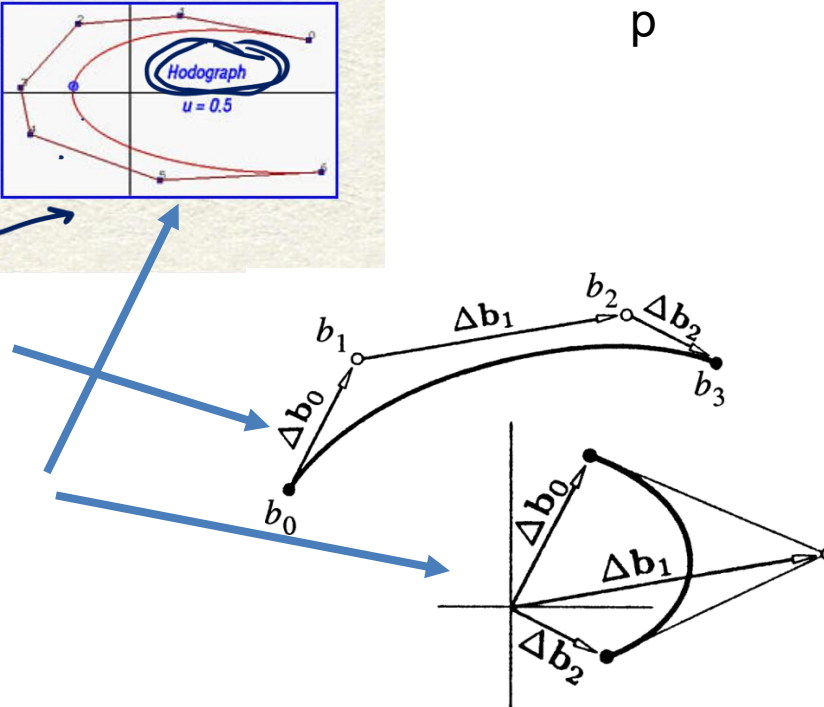
$$\frac{\partial}{\partial u} \sum_{i=0}^n B_{n,i}(u)P_i = n \left(\sum_{i=0}^{n-1} B_{n-1,i}(u)(P_{i+1} - P_i) \right)$$

Derivatives of Bezier Curve



Original Bezier Curve

Derivative Bezier Curve

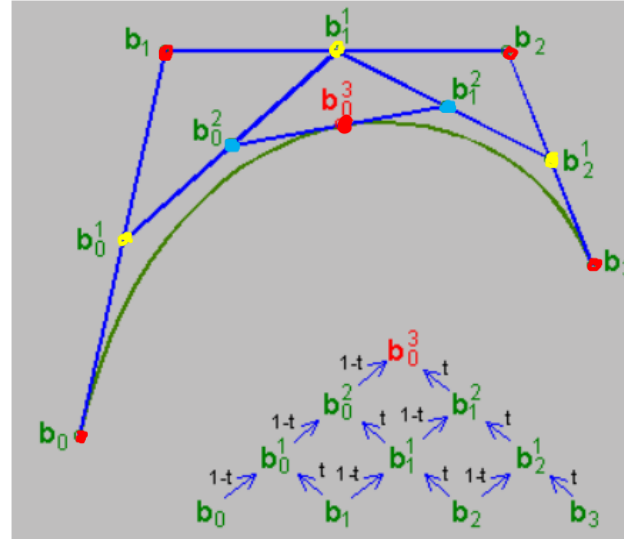
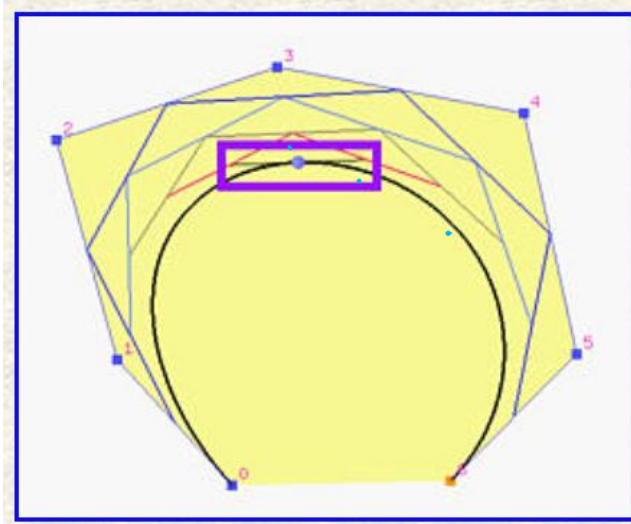


Subdivision of Bezier Curve (de Castljau Algorithm)



- Subdividing Bezier curve doesn't change its shape.
- De Casteljau's algorithm is a recursive method to evaluate Bézier curves. De Casteljau's algorithm can also be used to split a single Bézier curve into two Bézier curves at an arbitrary parameter value.
- **Steps:**
 1. Find the point on each edge of the control polygon at that subdivides the edge at specified value of u .
 2. Connect these points to form new set of segments
 3. Subdivide each of these at specified value of u
 4. Repeat this till only one segment can be created

Subdivision of Bezier Curve (de Casteljau Algorithm)



$u=0 \rightarrow 1$
 $u=0.25$
 $u=0.001$
 $u=0.002$

- The point at u value on the last segment is the same point on the Bezier curve (wrt original polygon) at same u which also subdivides the curve in same proportion.

Limitations of Bezier Curve



- Curve passes through only first and last control point and not through all control points thereby it is not exactly like interpolation. This can be inconvenient in some design situations
- Curve lacks local control. Every control point has effect on the shape of the curve over entire length. Selective editing of shape is not possible

An interesting blog about Bezier Curves:

<https://www.linearity.io/blog/bezier-curves/>

Practice Problems

Practice Problem - 1

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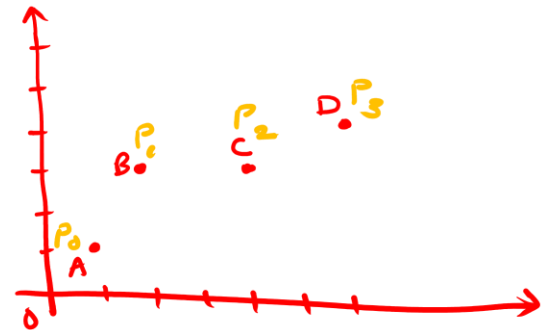
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Develop an equation of resulting Bezier curve for the coordinates of the four control points relative to a current WCS are given by A(1,1) B(2,3) C(4,3) and D(6,4), find the points on the curve and draw the plot for $u=0, 0.2, 0.4, 0.6, 0.8$ and 1.

Solution:

no. of control points = 4 $\therefore$ degree = $n-1 = 3$

$$\begin{aligned} A(1,1) &= P_0 & P_{0x} &= 1 & P_{0y} &= 1 \\ B(2,3) &= P_1 & P_{1x} &= 2 & P_{1y} &= 3 \\ C(4,3) &= P_2 & P_{2x} &= 4 & P_{2y} &= 3 \\ D(6,4) &= P_3 & P_{3x} &= 6 & P_{3y} &= 4 \end{aligned}$$



Practice Problem - 1

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Solution:

Parametric Equation for Bezier Curve:

$$P(u) = P_0 \cdot (1-u)^3 + P_1 \cdot 3 \cdot u \cdot (1-u)^2 + P_2 \cdot 3 \cdot u^2 \cdot (1-u) + P_3 \cdot u^3$$

$$x(u) = P_{0x} \cdot \frac{1 \cdot (1-u)^3}{1 \cdot (1-u)^3} + P_{1x} \cdot \frac{2 \times 3 \times u (1-u)^2}{2 \times 3 \times u (1-u)^2} + P_{2x} \cdot \frac{4 \times 3 \times u^2 \cdot (1-u)}{4 \times 3 \times u^2 \cdot (1-u)} + P_{3x} \cdot \frac{6 u^3}{6 u^3}$$

$$= (1-u)^3 + 6u(1-u)^2 + 12 \cdot u^2(1-u) + 6u^3 \quad \text{--- (I)}$$

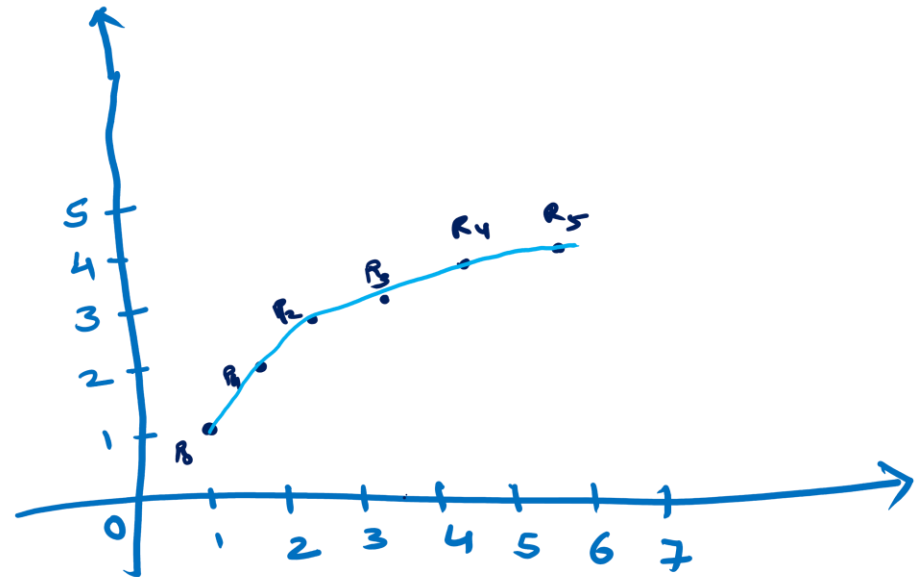
$$y(u) = P_{0y} \cdot \frac{1 \times (1-u)^3}{1 \times (1-u)^3} + P_{1y} \cdot \frac{3 \times 3 \times u \times (1-u)^2}{3 \times 3 \times u \times (1-u)^2} + P_{2y} \cdot \frac{3 \times 3 \times u^2 \times (1-u)}{3 \times 3 \times u^2 \times (1-u)} + P_{3y} \cdot \frac{4 \times u^3}{4 \times u^3}$$
$$= (1-u)^3 + 9u \cdot (1-u)^2 + 9u^2(1-u) + 4u^3 \quad \text{--- (II)}$$

Practice Problem - 1



Solution: $\text{eq}^n \text{I}$ $\text{eq}^n \text{II}$

	u	$x(u)$	$y(u)$
R_0	0	1	1
R_1	0.2	1.712	1.984
R_2	0.4	2.616	2.632
R_3	0.6	3.664	3.088
R_4	0.8	4.808	3.496
R_5	1	6	4



Practice Problem - 1



Solution:

Practice Problem - 2

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A cubic Bezier curve is defined by the control points as $(20, 20)$, $(60, 80)$, $(120, 100)$ and $(150, 30)$. Find the equation of curve and its midpoint.

Solution:

$$P(u) = P_0(1-u)^3 + P_1 \cdot 3 \cdot u \cdot (1-u)^2 + P_2 \cdot 3 \cdot u^2 \cdot (1-u) + P_3 \cdot u^3$$
$$x(u) = 20(1-u)^3 + 180 \cdot u \cdot (1-u)^2 + 360u^2(1-u) + 150 \cdot u^3$$
$$y(u) = 20(1-u)^3 + 240 \cdot u(1-u)^2 + 300u^2(1-u) + 30u^3$$

To find the coordinates of mid point (ie. $u=0.5$)

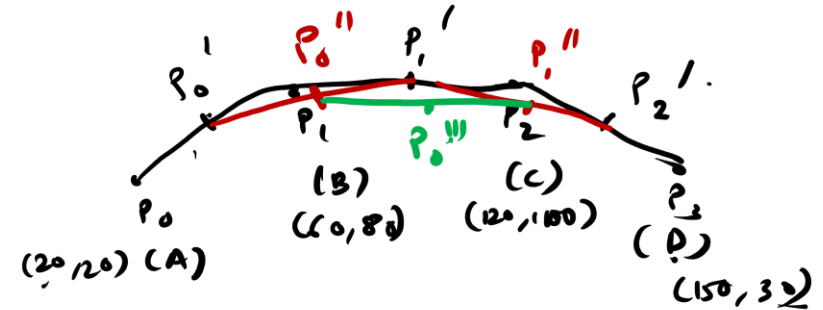
$$\left. \begin{aligned} x(u=0.5) &= 98.75 \\ y(u=0.5) &= 73.75 \end{aligned} \right\} \leftarrow \text{mid point of the curve}$$



Practice Problem - 2



Solution:



$$x = \frac{x_1 + x_2}{2}, \quad y = \frac{y_1 + y_2}{2}$$

Practice Problem - 2



Solution:

Practice Problem - 2



Solution:

Practice Problem - 3



- A]. Given a bezier curve with 4 control points- $B_0[1 \ 0]$, $B_1[3 \ 3]$, $B_2[6 \ 3]$, $B_3[8 \ 1]$. Determine points lying on the curve at $u = .25, 0.5$ and 0.75 . Also, draw a rough sketch of the curve.
- B]. With same control points as above, find the mid point of the Bezier curve without using its parametric equation

Solution:

Practice Problem - 2



Solution:

Practice Problem - 2



Solution:

Practice Problem - 2



Solution:

Thank You